

MODULE 10 · DOMAIN 2 - IMPLEMENTATION AND INTEGRATION

Fitting Into the Enterprise

Tasks 2.3 and 2.5. Event-driven integration, identity federation, hybrid and edge, CI/CD and the GenAI gateway.

DOMAIN 2 of 5 EXAM WEIGHT 26% MODULE 10 of 22

In a real enterprise deployment the model is perhaps five per cent of the work. The other ninety-five is **reaching systems that were built before it existed.**

Call it directly, or publish an event?

SYNCHRONOUS INTEGRATION

Your system calls theirs

- Simple, and the caller gets an answer
- Failures propagate straight back
- You are coupled to their availability
- Their rate limits become your rate limits

EVENT-DRIVEN

Something happened; react to it

- Producer and consumer never meet
- Add consumers without touching the producer
- Buffering absorbs spikes on both sides
- Eventual consistency, and harder tracing

THE TIEBREAKER

The exam guide says event-driven architectures to implement loose coupling, almost verbatim. When a scenario mentions a legacy system that must not be modified, that is the signal.

Adding AI without opening the old system

- ◆ **EventBridge** carries the event. New AI consumers subscribe without the source system knowing they exist.
- ◆ **Lambda webhook handlers** receive callbacks from SaaS systems that can push but cannot be extended.
- ◆ **API Gateway** fronts the AI capability as a microservice the legacy system can call with one HTTP request.
- ◆ **Change data capture** from DynamoDB Streams or a database, when the system emits nothing at all.
- ◆ Modifying the legacy system is the expensive answer, and usually the one the scenario has already excluded.

ON THE EXAM

The side-car is the shape to recognise: AI capability alongside the existing system, connected by events or one thin API, with the original left as it is.

An insurer runs a claims platform built in 2011. It writes every claim status change to an Oracle database and cannot be modified: the vendor no longer supports customisation and the internal team that wrote it has gone.

The business wants AI-generated summaries attached to claims whenever a claim reaches the "under review" status, and expects to add two more AI capabilities over the next year. Summaries do not need to appear instantly, but must appear reliably.

How should the AI capability be integrated?

- ☐ A Modify the claims platform to call an API Gateway endpoint whenever a status changes
- ☐ B Capture status changes from the database, publish them to Amazon EventBridge, and have AI consumers subscribe to the events they care about
- ☐ C Run a nightly job that queries the database for status changes and processes them in batch
- ☐ D Replatform the claims system so AI capabilities can be built in natively

◆ PAUSE AND DECIDE

How should the AI capability be integrated?

A Modify the platform to call an API
The scenario states it cannot be modified. The vendor does not support customisation and nobody remains who could do it.

C Nightly batch job
Works for the first capability and hard-couples every future one to a nightly cadence. It also polls rather than reacts.

B Change capture into EventBridge, consumers subscribe
Reads a source the platform already writes, without touching it, and the two future capabilities subscribe to the same events with no further integration work.

D Replatform the system
Replacing a claims platform to add a summary feature is the most expensive answer available, and would take longer than the year of roadmap described.

THE SIGNAL

"Expects to add two more AI capabilities over the next year" is the clause that selects an event bus. One consumer needs no bus; three consumers and counting do.

Identity is where enterprise AI integration usually breaks. The model needs to know **who is asking**, and most architectures lose that on the first hop.

Three ways enterprise identity reaches AWS

- ◆ **IAM Identity Center** federates the corporate directory into AWS access.
- ◆ **SAML 2.0 or OIDC** directly to an external IdP: Entra ID, Okta, Ping.
- ◆ **Amazon Cognito** for customer-facing identity, issuing tokens your API validates.
- ◆ Directory groups become **IAM role assumptions**, so department maps to permitted models.
- ◆ **AgentCore Identity** is the outbound half: acting on a user's behalf elsewhere.

ON THE EXAM

"Authenticate through Entra ID" means SAML or OIDC federation. Identity Center is the managed front end; direct federation answers when it is not in play.

There are **two** access decisions, not one.
May this user call this model? And may this
user see this data? Different mechanisms
answer each.

Carrying who is asking across accounts

- ◆ An agent assuming one shared role in each target account produces an audit trail naming **the role, not the person**.
- ◆ **sts:AssumeRole with session tags** passes user identity into the temporary credentials themselves.
- ◆ Those tags surface as **aws:PrincipalTag** in the target account, usable in the resource policy as an ABAC condition.
- ◆ They also appear in **CloudTrail**, so each action records which user's context performed it.
- ◆ Mark tags **transitive** to survive role chaining. Note that chained sessions cap at one hour.

ON THE EXAM

"Show which user's credentials performed each action" is the phrase. A shared role cannot answer it, and one role per user does not scale past a few dozen.

A company runs 25 AWS accounts under AWS Organizations. An assistant built on Amazon Bedrock AgentCore must query DynamoDB tables holding customer data, and each business unit keeps its tables in its own account.

Security requires strict data boundaries between business units, and a full audit trail showing which user's credentials performed each action. The assistant serves 50,000 interactions a day, so a design that scales with account count is essential.

How should the assistant access data across the accounts?

- ☐ A Give the assistant one IAM role per business unit account and assume the appropriate role for each query
- ☐ B Assume a role in the target account using `sts:AssumeRole` with session tags carrying the end user's identity, and condition the target resource policies on those tags
- ☐ C Create one IAM role per end user in every business unit account
- ☐ D Replicate all business unit DynamoDB tables into the assistant's own account and query them locally

How should the assistant access data across the accounts?

A One role per business unit account

Crosses the account boundary correctly and loses the user. Every action is logged as the shared role, so the audit requirement fails.

C One IAM role per end user per account

Preserves identity by brute force and does not scale. Roles per user multiplied across 25 accounts hits account limits and becomes unmanageable.

B AssumeRole with session tags carrying user identity

Preserves the boundary, carries user context into the credentials, and surfaces it in CloudTrail and in resource policy conditions.

D Replicate the tables centrally

Destroys the data boundary the requirement exists to protect, and adds replication lag and cost for every business unit.

THE SIGNAL

"Which user's credentials performed each action" is the deciding phrase. A shared role names itself in CloudTrail; only session tags put a person into that record.

What an IAM policy for Bedrock should say

- ◆ **Name the models.** Grant *bedrock:InvokeModel* on specific model or inference profile ARNs, not on a wildcard.
- ◆ Streaming is a **separate action**. *InvokeModelWithResponseStream* is not covered by *InvokeModel*.
- ◆ **Separate roles per application**, so CloudTrail attributes every call and one team's breach is contained.
- ◆ **SCPs** for organisation-wide boundaries, such as denying non-EU inference profiles, as module 1 showed.
- ◆ A single shared role with *bedrock:** makes attribution impossible and every guardrail optional.

ON THE EXAM

Granting on specific inference profile ARNs is how residency, cost control and model governance are all enforced at once, in a place no application code can bypass.

A consultancy is building an internal assistant over its document repository. Consultants belong to practice groups in the corporate directory, and a consultant may only see documents belonging to their own practice group and to the firm-wide library.

The application authenticates users through the corporate identity provider, then calls Bedrock and a knowledge base using a single Lambda execution role. Practice group is currently passed from the browser as a request field.

What should the consultancy change?

- ☐ A Validate the practice group field against the list of valid practice groups before using it as a filter
- ☐ B Federate the directory through IAM Identity Center, derive the practice group from the verified token in the backend, and apply it as the retrieval filter
- ☐ C Create one Lambda execution role per practice group, each with its own Bedrock permissions
- ☐ D Build a separate knowledge base for each practice group

◆ PAUSE AND DECIDE

What should the consultancy change?

A Validate the field against valid groups
Module 4's lesson again. Every other practice group is a valid value, so validation confirms the format while leaving the caller free to choose.

C A role per practice group
IAM controls model access, not which chunks a retrieval returns. A role cannot express "this practice group plus the firm-wide library".

B Federate, derive from the token, filter server-side
The claim comes from the identity provider rather than the browser, so the user cannot choose their own filter, and the corporate directory stays the source of truth.

D A knowledge base per group
Works, and duplicates the firm-wide library into every one of them, multiplying ingestion, cost and drift for a filter that would do.

THE SIGNAL

Model access and data access are different problems, and C answers the wrong one. IAM decides whether you may call Bedrock; a metadata filter decides what comes back.

There is no on-premises Bedrock. AWS does not ship a self-hosted version, so **every hybrid pattern is about what crosses the boundary**, not about moving the model.

Keep the data local, or keep everything local?

DATA LOCAL, INFERENCE IN REGION

Retrieval local, prompt travels

- Vector store and documents stay on Outposts
- Only the assembled prompt leaves
- PrivateLink or Direct Connect for the path
- Full Bedrock catalogue still available

FULLY LOCAL

Model and data both local

- Open-weight model on SageMaker on Outposts
- Nothing leaves the building
- You operate the GPUs and the model
- No Bedrock, so no Guardrails or Flows

THE TIEBREAKER

Read the requirement precisely. "Documents must not leave" allows the first. "No data of any kind may leave" forces the second, and forces you off Bedrock.

What each hybrid service is actually for

SERVICE	WHERE IT SITS	THE REASON TO USE IT
AWS Outposts	Your data centre	Data residency and local system latency
AWS Local Zones	A metropolitan area	Low latency to a city, without on-premises hardware
AWS Wavelength	Inside a 5G network	Ultra-low latency to mobile devices
Direct Connect	A private circuit	Dedicated bandwidth between site and Region
PrivateLink	Inside the AWS network	Reaching a service without traversing the internet

ON THE EXAM

Wavelength is about mobile devices, not data residency. If a scenario mentions a regulator rather than a handset, Wavelength is wrong however hybrid it sounds.

A defence contractor must build a document assistant over classified engineering drawings. Its security policy states that no drawing, no extract of a drawing, and no text derived from one may leave its facility under any circumstances.

The facility already runs AWS Outposts. The team has proposed keeping the vector store on the Outpost and sending only retrieved passages to Amazon Bedrock in the Region over AWS Direct Connect.

Does the proposal meet the requirement, and if not, what does?

- ☐ A Yes. Direct Connect is a private circuit, so the passages never traverse the public internet
- ☐ B No. Retrieved passages are text derived from the drawings. Host an open-weight model on SageMaker AI on the Outpost so inference stays local
- ☐ C No. Use an EU geographic inference profile so processing stays within the correct jurisdiction
- ☐ D Yes, provided Bedrock model invocation logging is disabled so no derived text is retained

◆ PAUSE AND DECIDE

Does the proposal meet the requirement, and if not, what does?

- A

 Yes, Direct Connect is private
Confuses a private path with data not leaving. The passages still travel to a Region and are processed there, which the policy forbids outright.
- B

 No. Open-weight model on SageMaker on the Outpost
"No text derived from one may leave" includes retrieved passages, so inference has to happen locally. That means giving up Bedrock and hosting the model yourself.
- C

 EU geographic inference profile
Bounds processing to a geography. The requirement is a facility, not a geography, and a geographic profile still means a Region.
- D

 Disable invocation logging
Changes what is retained, not what is transmitted and processed. The data still left the facility before anyone logged anything.

THE SIGNAL

"No text derived from one" is doing all the work. Retrieved chunks are derived text, so the entire retrieval-in-Region pattern is excluded and Bedrock goes with it.

A GenAI pipeline has a problem ordinary CI/CD does not: **the tests are non-deterministic**, so a green build is a probability rather than a proof.

Four artefacts, not one

- ◆ **Application code.** The ordinary part, and the only part most pipelines handle.
- ◆ **Prompt versions.** Immutable, and promoted through the pipeline like any release.
- ◆ **Model selection.** In AppConfig or a prompt version, so it changes without a deployment.
- ◆ **The evaluation dataset.** Change the golden set and scores stop being comparable.
- ◆ **Infrastructure** via CDK, including knowledge base and guardrail configuration.

ON THE EXAM

Guardrail configuration belongs in infrastructure as code for the same reason security groups do. One edited by hand is a difference nobody can see.

Getting a GenAI change to production

01

Build

CodeBuild compiles, plus security and dependency scans.

02

Evaluate

Run the golden set. Record a score, not a pass.

03

Gate

Score below threshold stops the release automatically.

04

Canary

Deploy to a traffic slice and watch quality metrics.

05

Roll back

Repoint at the previous prompt version and model.

ON THE EXAM

Rollback here is cheap for the same reason it was in modules 2 and 6: the previous versions still exist as artefacts. You repoint rather than rebuild.

A retailer deploys its AI assistant through AWS CodePipeline. The pipeline builds the Lambda code, runs unit tests, and deploys through CloudFormation. Prompts are stored in Bedrock Prompt Management and the model ID sits in AWS AppConfig.

Last week a prompt version was published directly in the console and quality dropped for two days before anyone noticed. The pipeline was green throughout, because nothing about the application code had changed. The team wants prompt changes governed as strictly as code changes, without slowing routine releases.

What should the team add?

- ☐ A Restrict console access to Prompt Management so only the pipeline can publish versions, and add an evaluation stage that scores a golden set and gates promotion on the result
- ☐ B Add unit tests that assert the prompt text matches an expected string
- ☐ C Move the prompts back into the Lambda source code so they are covered by the existing pipeline
- ☐ D Enable CloudTrail on Prompt Management so console publications are recorded

What should the team add?

- A

Restrict publishing to the pipeline, plus an evaluation gate

Closes the bypass and adds the quality signal the pipeline lacked. Both halves are needed: one stops the shortcut, the other catches a bad prompt that goes through it.
- B

Assert the prompt text in unit tests

Tests that the prompt is what you wrote, not that it works. Any deliberate change updates the assertion and the test passes again.
- C

Prompts back into source code

Fixes the governance gap by discarding versioning, the approval gate and model binding. Module 6 explained why prompts left the codebase.
- D

CloudTrail on Prompt Management

Detective, not preventive, and detection was never the failure. Two days passed with a full audit trail available and nobody looking at it.

THE SIGNAL

Two problems: a path that bypassed the pipeline, and a pipeline with no quality signal for prompts. The answer must close both, which is why only A works.

Making the capability easy to consume

- ◆ **OpenAPI specifications** for API-first development, so consumers integrate against a contract before it is built.
- ◆ **AWS Amplify** for declarative front-end components, when the interface matters more than the backend.
- ◆ **Bedrock Flows** as a no-code workflow builder for teams who are not engineers.
- ◆ The **GenAI gateway** from module 8 is the same idea at platform scale: one contract, many consumers.
- ◆ A capability nobody can adopt without a two-week integration is a capability most teams will not adopt.

ON THE EXAM

Skill 2.5.2 is about adoption, which is an unusual thing for an exam to test. The signal is a scenario where the AI works and nobody is using it.

Developer tooling, and a name change to know

- ◆ **Amazon Q Developer** IDE plugins reach end of support on **30 April 2027**. New signups closed on 15 May 2026.
- ◆ **Kiro** is the named successor: a spec-driven agentic IDE, GA November 2025. Both are on the in-scope list.
- ◆ Q Developer in the **AWS Console** and first-party AWS experiences are **not** affected by that sunset.
- ◆ For skill 2.5.6, the troubleshooting tools are **CloudWatch Logs Insights** for prompts and responses, and **X-Ray** for tracing FM calls.
- ◆ On the exam, Q Developer remains a correct answer for code generation and refactoring.

ON THE EXAM

The fourth case of exam drift in this course, and the same shape as Kendra. The guide names the retiring product and its successor side by side, so know both.

A logistics company wants AI-assisted intake for paper delivery notes: scan the note, extract eight fields, validate them against the order system, route exceptions to a human, and write the result into an existing warehouse management system that exposes a REST API.

The process must be auditable end to end, must retry failed steps, and must pause for human review on exceptions. An engineer has proposed one large Lambda function that performs every step in sequence.

How should this be built?

- ☐ A The single Lambda function as proposed, with try-except blocks around each step
- ☐ B AWS Step Functions orchestrating Bedrock Data Automation for extraction, Lambda for validation and the warehouse API call, and a task token callback for human review
- ☐ C A Bedrock agent with tools for extraction, validation and the warehouse API
- ☐ D Amazon Bedrock Flows, with a condition node routing exceptions

◆ PAUSE AND DECIDE

How should this be built?

- A

 One large Lambda with try-except
Cannot pause for human review inside a fifteen-minute execution, and every retry re-runs steps that already succeeded. Audit means reading logs.
- B

 Step Functions with BDA and a task token callback
The sequence is known, so it is a workflow. Execution history is the audit trail, per-state retries are built in, and the task token pauses for the human.
- C

 A Bedrock agent with tools
Module 7's rule. The steps are fixed and listed in order, so making them non-deterministic buys flexibility nobody asked for.
- D

 Bedrock Flows with a condition node
Handles the AI steps and branching well. A human pause of unbounded length and a call into a legacy REST API are outside what it is for.

THE SIGNAL

Three requirements name Step Functions features directly: end-to-end audit is execution history, retry failed steps is per-state retry, and pause for human review is the task token callback.

The whole domain on one slide

- ◆ **Agents** for unknown paths. If you can draw the flowchart, build the flowchart.
- ◆ **The API layer** follows one question: who is waiting, and for how long?
- ◆ **Deployment** defaults to Bedrock. Self-host for a model it lacks, weights you own, or control you need.
- ◆ **Integration** disturbs the existing system as little as possible. Events over modification.
- ◆ **Controls belong where they cannot be argued with:** IAM, the tool, the pipeline gate. Not the prompt.

ON THE EXAM

That is 26% of the exam, and 57% now covered with Domain 1. Module 11 opens Domain 3 with guardrails and content safety.